OCS SYSTEM ADAPTATION IN SONIC-MGMT FRAMEWORK

An approach for validating OCS system in sonic-mgmt

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OVERVIEW

- OCS characteristics
- How integrating OCS into the sonic-mgmt framework
- Physical topology
- OCS VLAN-based verification strategy
- Roadmap

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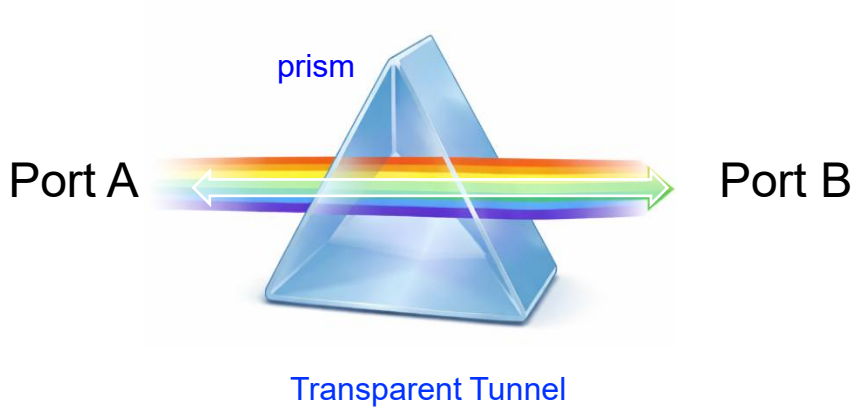
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OCS CHARACTERISTICS

OCS : Layer0, Physics, Paths

Core Concept:

- Steering traffic directly in optical domain
- No routing logic
- No packet forwarding tables (like FDB)
- Transparent tunnel (Exclusive physical path)



Key Points:

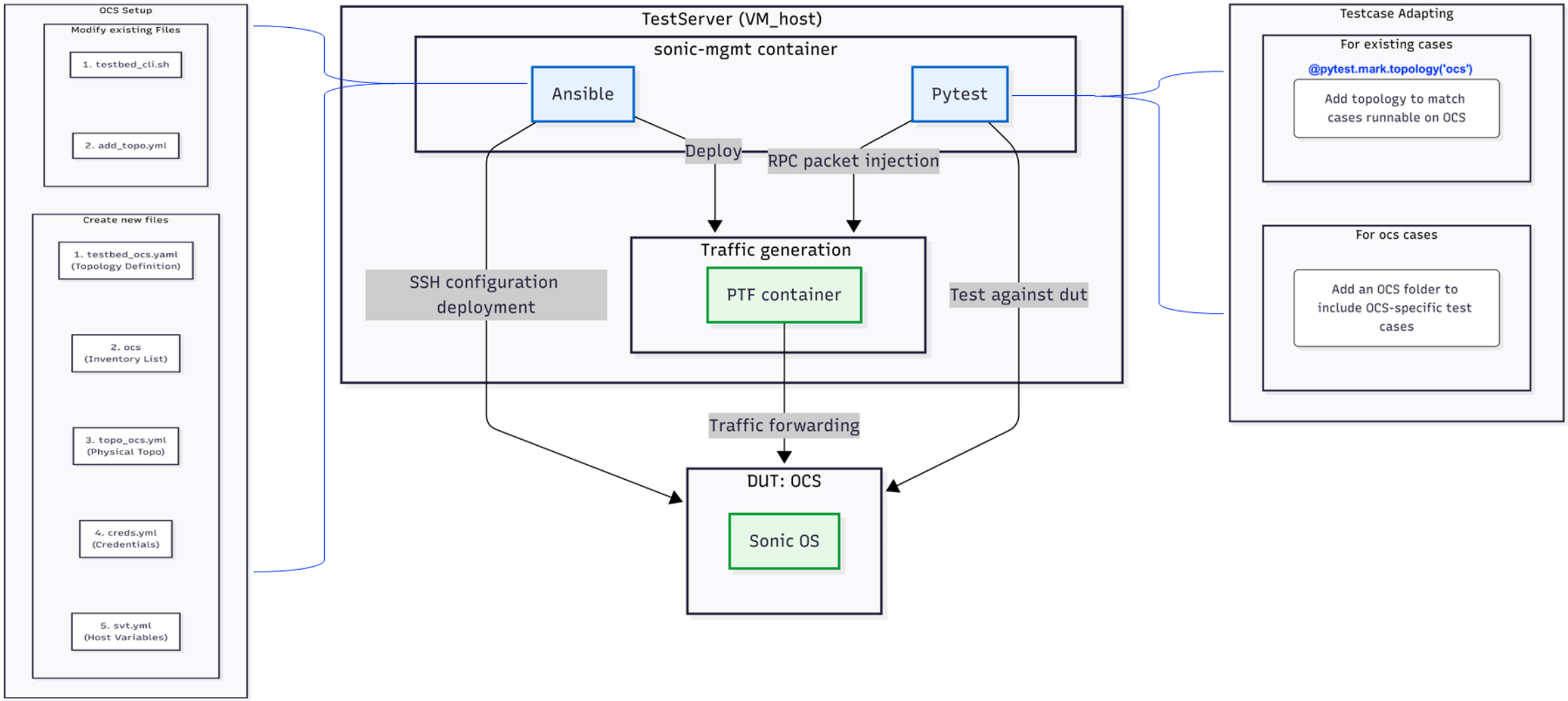
- Optical path
- Packet-format/speed transparency
- Port level dynamic path
- Blind to traffic

? The Challenge:
How to leverage sonic-mgmt (Packet Logic) to test OCS (Physical Paths)

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HOW – INTEGRATING OCS INTO THE SONIC-MGMT FRAMEWORK



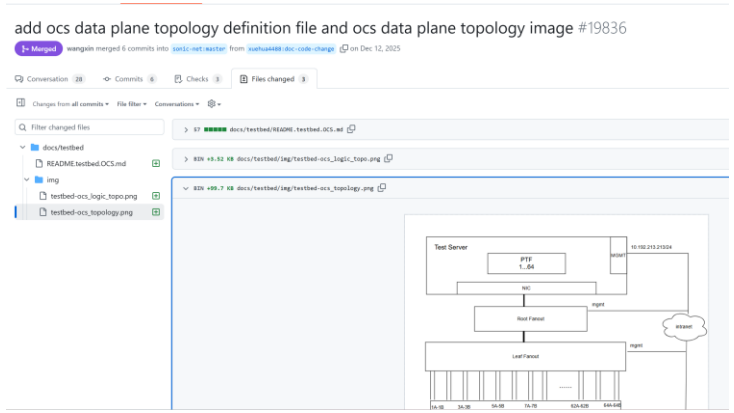
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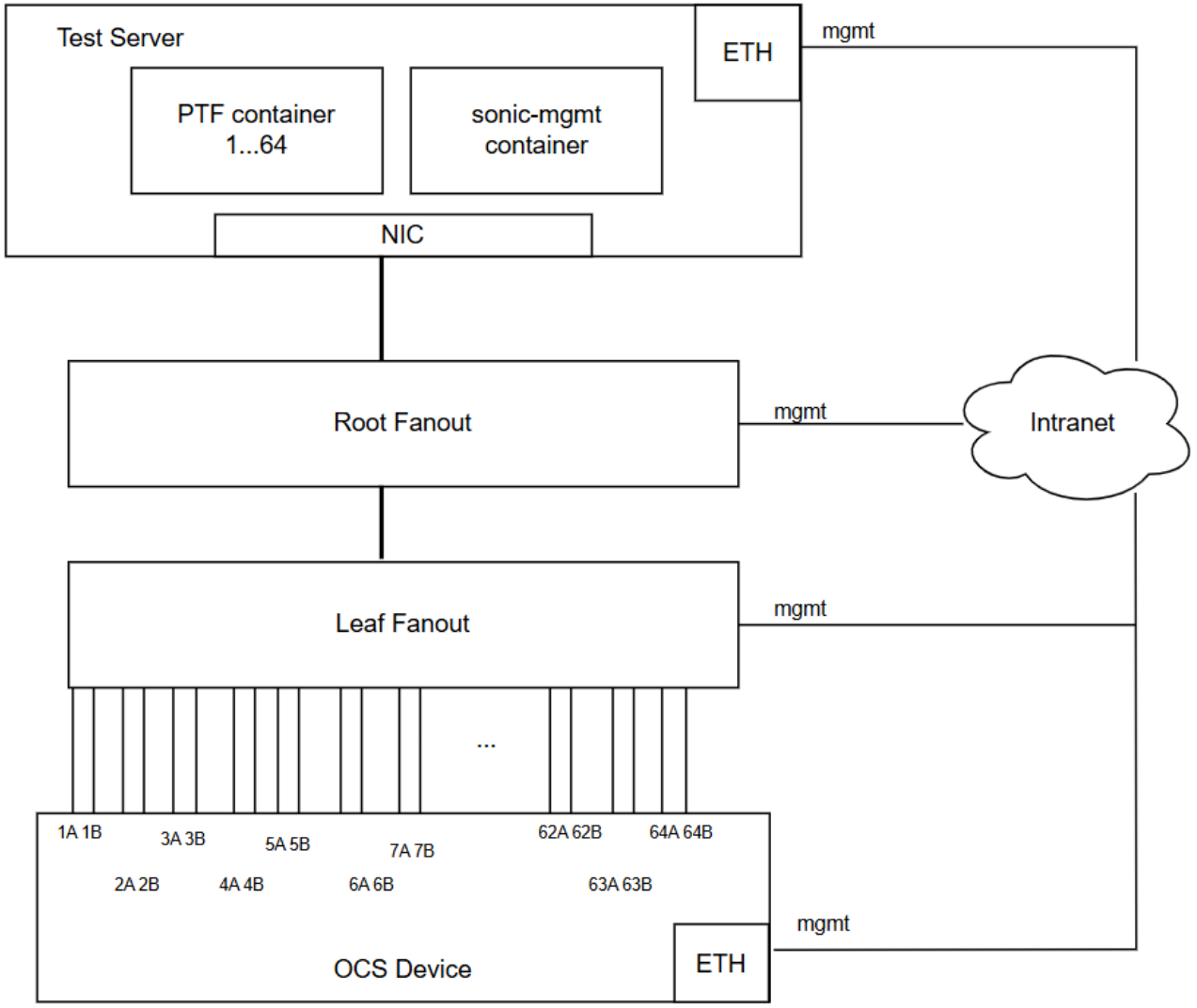
PHYSICAL TOPOLOGY

Key components :

- Test servers
- Fanout switches
  - Root fanout switch
  - Leaf fanout switch
- OCS device

Approved PR#19836





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OCS VERIFICATION STRATEGY

Solution – VLAN-based Abstraction:

- Mapping physical light paths to logical VLAN tags.

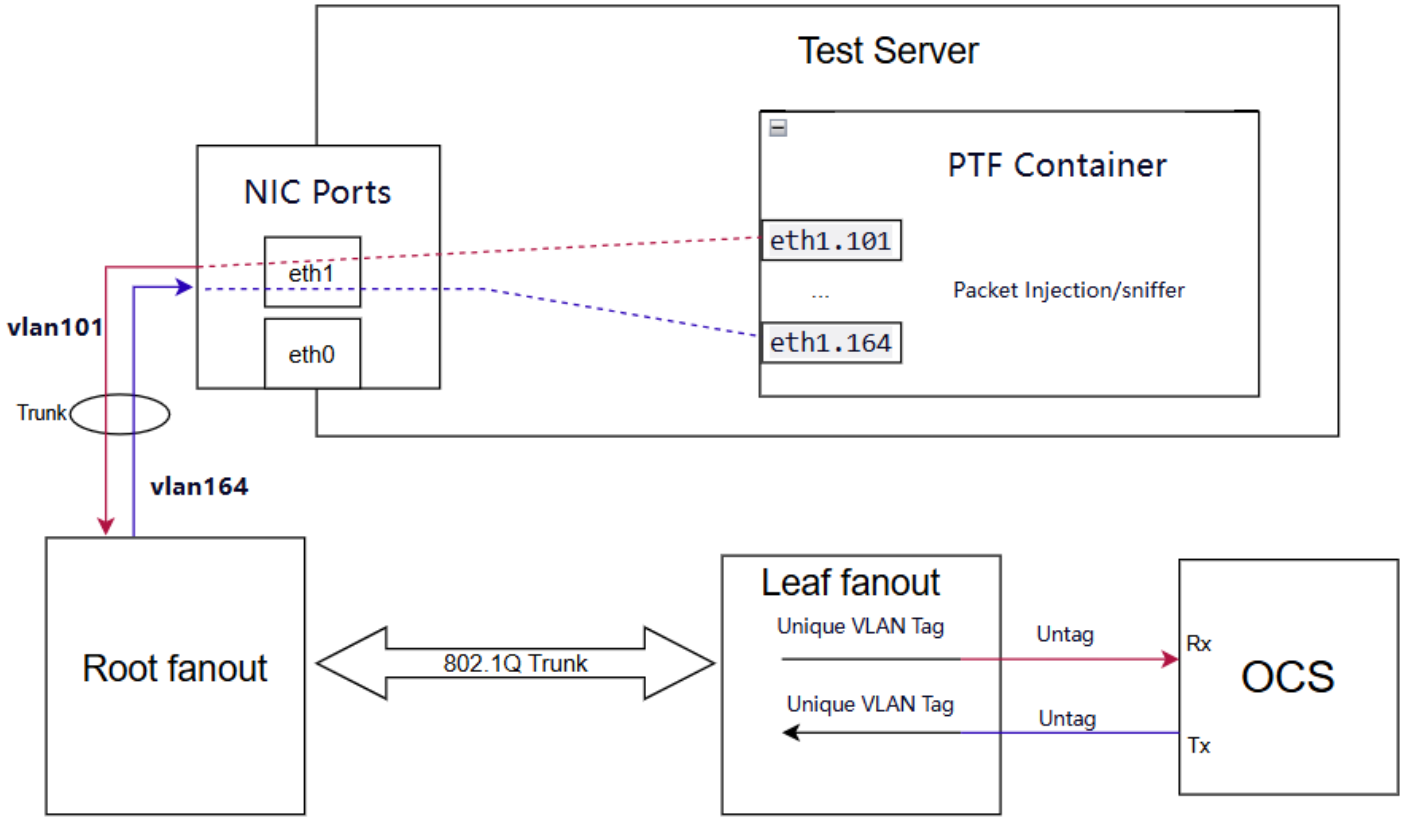
Verification & Status

- config ocs cross-connect add 2A-64B
- show ocs cross-connect status

```
root@sonic:~# show ocs cross-connect status
id      a_side    b_side    a_side_status    b_side_status
------  --------  --------  ---------------  ---------------
2A-64B  2A        64B       tuned            tuned
```

Traffic Flow Logic:

- **Leaf Fanout:**
  - Assigns a unique VLAN tag per OCS port (Rx/Tx).
- **Root Fanout:**
  - Aggregates tagged streams via 802.1Q trunks.



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ROADMAP FOR OCS ADAPTATION IN SONIC-MGMT FRAMEWORK

- **Phase 1: Environment Adaptation**
  - Estimated duration : 1 Month
  - Status: Ongoing
  - Objective: Make OCS recognized as a valid DUT (Device Under Test) within the SONiC-mgmt ecosystem
  - Scope: Topology definition, testbed setup definition, and basic deployment changes required for OCS

- **Phase 2: Test Case Migration & Integration**
  - Estimated duration : 2~3 Months
  - Status: Not Started
  - Objective: Reuse existing SONiC-mgmt code as much as possible and add OCS-specific wrappers only when needed.

- **Phase 3: Execution and Validation**
  - Estimated duration : 4~8 Months
  - Status: Not Started
  - Objective: Integrate OCS test cases into sonic-mgmt regression and CI flow.

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CURRENT PROGRESS

Phase 1: Environment Adaptation

Definition files have been merged to the *SONiC-mgmt* master branch

- OCS topology definition *:* _OCS Topology Definition
- Testbed setup definition *:* OCS Setup
- HLD document: OCS.HLD.md

Deployment Modification (Code Changes): **Ongoing**

ID	Folder	File	Status
1	ansible/	add file testbed_ocs.yaml	merged
2	ansible/	add file config_ptf_network_for_ocs.sh	merged
3	ansible/var	add file topo-ocs.yml	merged
4	ansible/host_var	add file creds.yml	merged
5	ansible/group-var/ocs	add file svt.yml	Ongoing https://github.com/sonic-net/sonic-mgmt/pull/25692
6	ansible/templates/	add file config_db.ison.i2	Plan
7	ansible/	add file: ocs, config_ocs.yml	Plan
8	ansible/roles/vm _set/tasks	modify add-topo.yml	Plan
9	ansible/	modify testbed-cli.sh	Plan

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PHASE 2 PLAN

Phase 2: Test Case Migration & Integration

- Command Reuse Verification:
  - Validate standard duthost commands on OCS, such as show interface and config aaa.
  - Identify OCS-specific CLI differences, such as config ocs cross-connect

- Fixture Adaptation:
  - Review existing SONiC-mgmt fixtures (e.g., duthost, tbinfo).
  - Add lightweight wrappers only when OCS behavior differs from standard switches.(e.g., ocs cross-connect command).

- Redis Database Alignment:
  - Ensure OCS populates standard SONiC CONF_DB, STATE_DB, etc., so existing verification logic can be reused.

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PHASE 3 PLAN

- **Phase 3: Execution and Validation**

- Initial E2E Validation Run:
  - Execute the first end-to-end test case on the new Testbed/Topology
  - Debug connectivity, timing, and stability issues

- OCS Test Case Integration:
  - Integrate OCS-related cases into the sonic-mgmt test suites.
  - Align test execution, reporting, and pass/fail criteria with sonic-mgmt

- Regression Suite Integration:
  - Group OCS tests into specific test groups (e.g., sanity, functional).
  - Integrate OCS tests into the run_tests.sh script execution flow

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